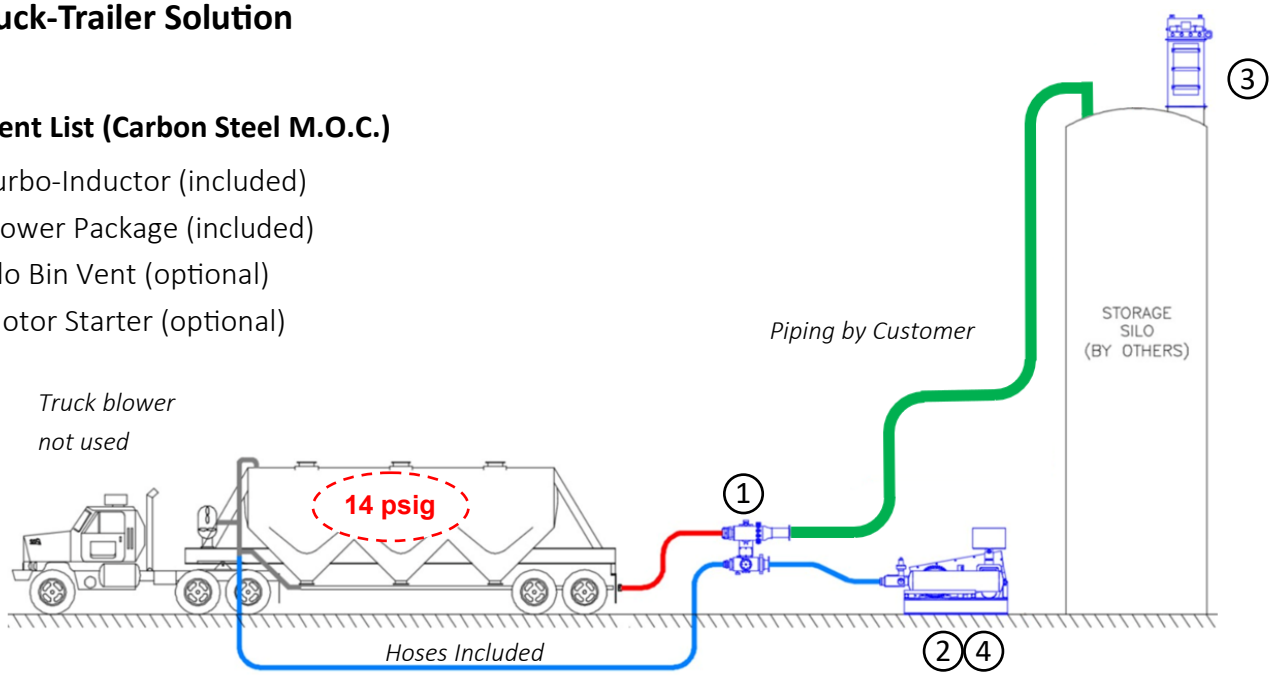
CYCLONAIRE CEMENTITIOUS MATERIALS APPLICATIONS[1]

Turbo-Inductor System Design Guide

P/D Truck-Trailer Solution

Equipment List (Carbon Steel M.O.C.)

1. (1) Turbo-Inductor (included)
2. (1) Blower Package (included)
3. (1) Silo Bin Vent (optional)
4. (1) Motor Starter (optional)



ESTIMATED TURBO-INDUCTOR CAPACITY (STPH)

PD Truck-Trailer Unloading							
	HOSE[2] →	Discharge Hose Size = 4”			Discharge Hose Size = 5”		
	Convey Line Size →	6”	8”	10”	6”	8”	10”
Equivalent Distance to silo[3] (ft)	210	38	48	54	43	57	66
	260	34	44	51	37	51	62
	310	30	41	49	32	46	58
	360	27	38	47	29	42	54
	410	25	36	44	26	39	51
	460	23	33	42	24	36	48
1	Turbo-Inductor	T-I 4/6	T-I 4/8	T-I 4/10	T-I 5/6	T-I 5/8	T-I 5/10
2	Blower SCFM[4]	780	1300	2050	780	1300	2050
3	Optional Bin Vent[5]	36-DC-25	36-DC-36	36-DC-64	36-DC-25	36-DC-36	36-DC-64

1. Class 1 materials: fine, free-flowing, fluidizable materials that retain aeration including cement, fly ash, and slag. Assumed material properties: ~94 lbm/ft3 (settled), 74-micron particle size (200 mesh).
2. Assumes a 15-foot conveying hose between trailer and T-I and a 4” 25-ft trailer discharge manifold (Sch. 40). Hose routing should be as straight as possible with minimal bends.
3. Downstream eq. pipe length, Turbo-Inductor to silo. (1) 90° elbow = 20 feet.
4. Performance based on 70 degrees F; blower output applicable to 2000 feet ASL. Estimated rates based on 14 psig in the trailer. Blowers are configured for 15 psig discharge pressure and include a pressure transmitter.
5. Bin vent dust collectors are static configuration (no fan) and include a differential pressure switch.

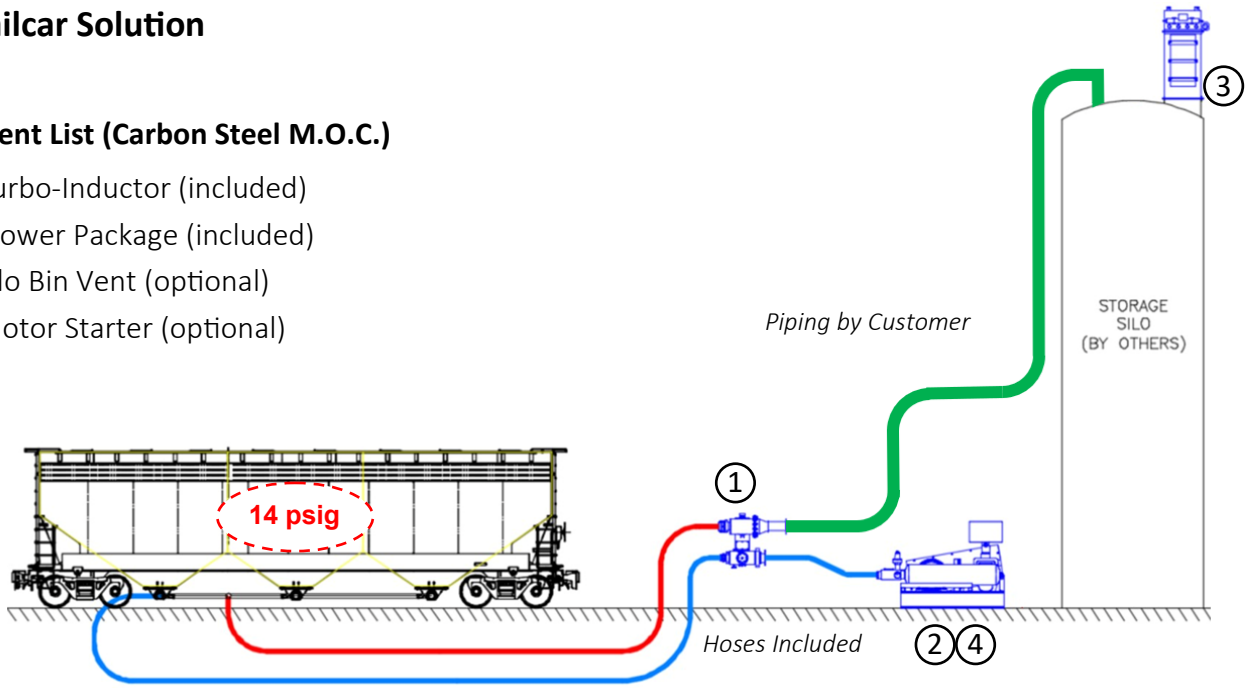
CYCLONAIRE CEMENTITIOUS MATERIALS APPLICATIONS[1]

Turbo-Inductor System Design Guide

P/D Railcar Solution

Equipment List (Carbon Steel M.O.C.)

1. (1) Turbo-Inductor (included)
2. (1) Blower Package (included)
3. (1) Silo Bin Vent (optional)
4. (1) Motor Starter (optional)



ESTIMATED TURBO-INDUCTOR CAPACITY (STPH)

PD Railcar Unloading						
HOSE[2] →		Discharge Hose Size = 5”			Discharge Hose Size = 6”	
Convey Line Size →		6”	8”	10”	8”	10”
Equivalent Distance to silo 3 (ft)	210	43	58	69	64	78
	260	37	52	64	56	71
	310	33	47	60	50	65
	360	29	43	56	46	60
	410	26	40	52	42	56
	460	24	37	49	38	52
1 Turbo-Inductor		T-I 5/6	T-I 5/8	T-I 5/10	T-I 6/8	T-I 6/10
2 Blower SCFM[4]		780	1300	2050	1300	2050
3 Optional Bin Vent[5]		36-DC-25	36-DC-36	36-DC-64	36-DC-36	36-DC-64

1. Class 1 materials: fine, free-flowing, fluidizable materials that retain aeration including cement, fly ash, and slag. Assumed material properties: ~94 lbm/ft3 (settled), 74-micron particle size (200 mesh).
2. Assumes a 15-foot conveying hose between railcar and T-I and a 5” 15-ft railcar discharge manifold (Sch. 40). Hose routing should be as straight as possible with minimal bends.
3. Downstream eq. pipe length, Turbo-Inductor to silo. (1) 90° elbow = 20 feet.
4. Performance based on 70 degrees F; blower output applicable to 2000 feet ASL. Estimated rates based on 14 psig in the railcar. Blowers are configured for 15 psig discharge pressure and include a pressure transmitter.
5. Bin vent dust collectors are static configuration (no fan) and include a differential pressure switch.